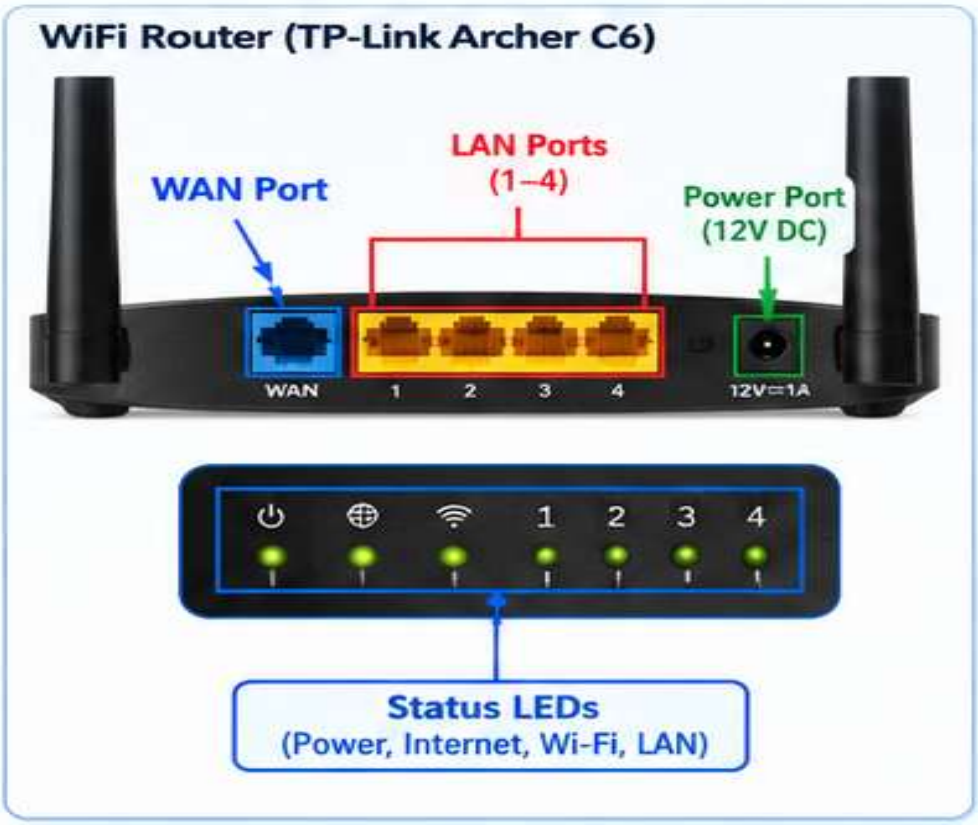
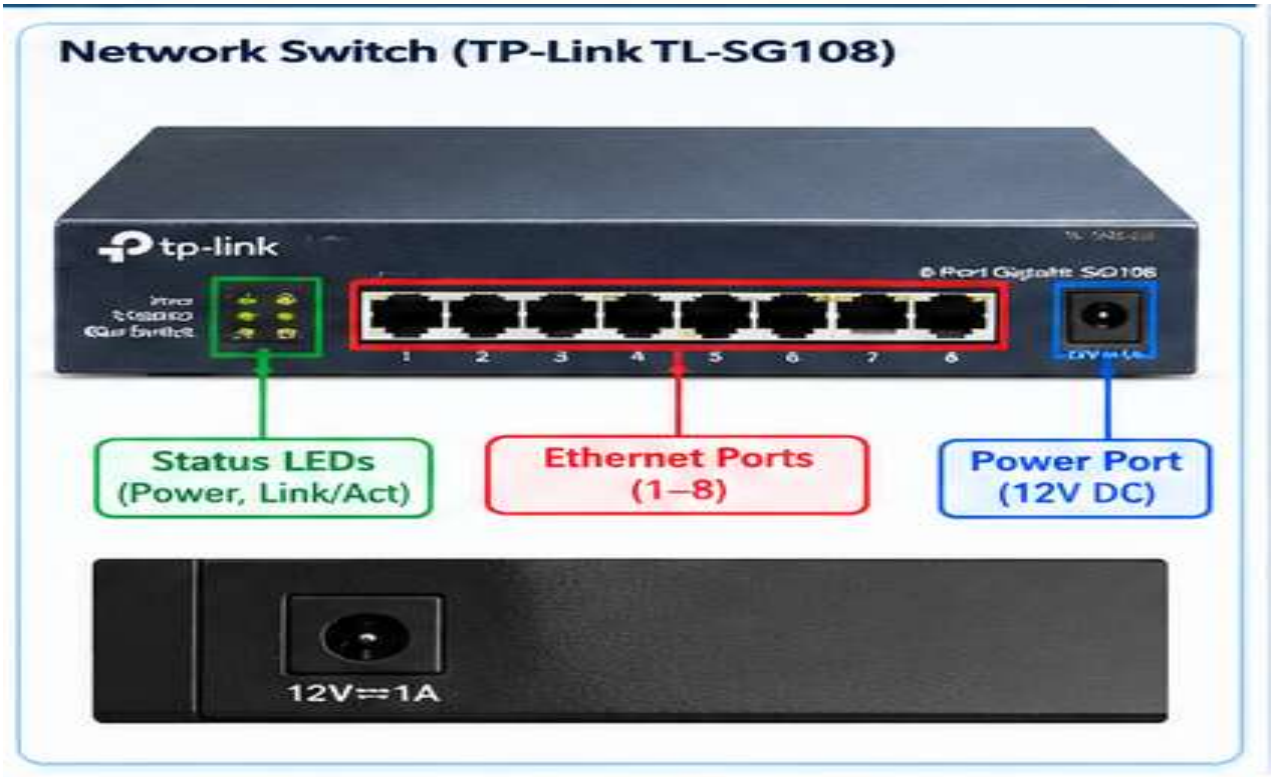
1. **Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.**





2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Feature	Router	Switch
Main function	Connects different networks, such as a home network and the internet.	Connects multiple devices within the same local network.
Address used	Uses IP addresses to route packets between networks.	A typical Layer 2 switch uses MAC addresses to forward Ethernet frames.
Network connection	Connects networks such as LAN and WAN.	Primarily connects devices within a LAN.

3. **Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.**

1. Wi-Fi Router (TP-Link Archer C6)



Power Port
Connects the router to the power adapter (12V DC).

WAN Port
Connects to the modem or ISP device.

LAN Ports (1–4)
Connect wired devices like computers, switches, etc.

WPS/Reset Button
Used for WPS pairing or factory reset.

Status LEDs
Show power, internet, Wi-Fi and LAN activity.

2. Network Switch (TP-Link TL-SG108)



Status LEDs
Show power and link/activity for each port.

Ethernet Ports (1–8)
Connect computers, routers, printers, etc. using Ethernet cables.

Power Port
Connects the switch to the power adapter.

Router – Key Ports & LEDs

- **Power Port:** 12V DC (connects power adapter)
- **WAN Port:** connects to modem/ISP
- **LAN Ports (1–4):** connect wired devices
- **WPS/Reset Button:** pairing or reset
- **Status LEDs:** power, internet, Wi-Fi, LAN activity

Switch – Key Ports & LEDs

- **Power Port:** 9V DC (connects power adapter)
- **Ethernet Ports** (1–8): connect wired devices
- **Status LEDs:** power and link/activity for each port

4. **Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.**

- Step 1: Connect the ISP internet line to the modem or ONT, if required by the ISP.
- Step 2: Connect the modem or ONT to the router's WAN port using an Ethernet cable.
- Step 3: Connect one LAN port of the router to a port on an eight-port or larger Gigabit Ethernet switch.
- Step 4: Connect all eight gaming PCs to the switch using separate Ethernet cables.
- Step 5: Configure the router's internet connection, DHCP and security settings. Test internet connectivity on every PC.
- Justification: The router must connect the café's local network to the internet and manage traffic between networks. The switch provides enough wired connections for all eight gaming PCs. Wired Ethernet connections offer stable, low-latency performance for multiplayer gaming and reliable IPL streaming.